Active ants: dynamics

$$\begin{cases} \partial_t f & = \sigma_x \Delta_x f - \mathrm{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) & = f_0 \end{cases}$$

Lyapunov-Schmidt method

- Linearise PDE around $f = 1/2\pi$: operator L

- Compute the kernel of L (i.e. at the bifurcation point)

- In our case $\dim(L) = 4$

- Project the PDE onto the basis of the kernel

- Lots of integrals, but doable

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- What happens as $t \to \infty$?